Akoizo Original Practice - Form 1 - Answer key

Division B | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which operation preserves the exact letter counts while changing order?

A. Transposition

B. Monoalphabetic letter replacement

C. Deleting spaces and letters

D. A numerical hash

Correct: A

2. (2 points) Which letter is decoded?

Unit: Porta, Nihilist, and checkerboard tables.
Worked-case data: Nihilist ciphertext: 65. Key coordinate: 23. Square entry: Row 4, column 2 contains R.
Separate investigation observations:
Nihilist ciphertext: 65.
Key coordinate: 23.
Square entry: Row 4, column 2 contains R.

A. R

B. The letter in row 2, column 3

C. No letter because 65 is outside the square

D. The letter in row 6, column 5

Correct: A

3. (2 points) Which handoff is most useful?

Unit: Competition strategy and deliberate practice.
Worked-case data: Puzzle A offers 100 points with a 0.9 estimated solve probability in 3 minutes; B offers 200 with 0.3 probability in 6 minutes. Compare expected points per minute as a planning aid.

A. State cipher, key direction, and confirmed mappings

B. Write only “almost solved”

C. Erase all tentative work

D. Give only the final guessed word

Correct: A

4. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Atbash and Caesar shifts.
Worked-case data: Cipher: Caesar with forward encryption shift 3. Ciphertext: FDW. Alphabet: A=0 through Z=25.
Separate investigation observations:
Cipher: Caesar with forward encryption shift 3.
Ciphertext: FDW.
Alphabet: A=0 through Z=25.

answer: Decode by shifting backward three places. F becomes C, D becomes A, and W becomes T. Re-encrypt CAT to check that it gives FDW. (4 points).

5. (2 points) What remains true in a xenocrypt substitution?

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

A. Every word is a cognate

B. Accents are always separate symbols

C. Mappings must be globally consistent

D. English frequency ranks are exact

Correct: C

6. (4 points) Explain “Caesar cipher” in your own words.

Unit: Atbash and Caesar shifts.
Worked-case data: Cipher: Caesar with forward encryption shift 3. Ciphertext: FDW. Alphabet: A=0 through Z=25.
Separate investigation observations:
Cipher: Caesar with forward encryption shift 3.
Ciphertext: FDW.
Alphabet: A=0 through Z=25.

answer: A fixed alphabet shift. (4 points).

7. (2 points) What must be known before using a Baconian table?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

A. The alphabet convention

B. The carrier sentence meaning only

C. The most common English word

D. The paper color

Correct: A

8. (4 points) Explain “Transposition” in your own words.

Unit: Columnar transposition and homophonic evidence.
Worked-case data: Rectangle: Six letters in two columns. Ciphertext pieces: CTO and ATO. Clue: The plaintext starts with CAT.
Separate investigation observations:
Rectangle: Six letters in two columns.
Ciphertext pieces: CTO and ATO.
Clue: The plaintext starts with CAT.

answer: Rearrangement of letters without changing their identities. (4 points).

9. (4 points) Give the pattern of LEVEL.

Unit: Monoalphabetic substitution and language evidence.
Worked-case data: A ciphertext word ABBC is proposed to mean MEET or TEAM. Which fits the equality pattern?

answer: 0-1-2-1-0. (4 points).

10. (4 points) Write the SEND + MORE units equation.

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

answer: D + E = Y + 10c1, with c1 the outgoing carry. (4 points).

11. (4 points) Why must an ordinary substitution table be one-to-one?

Unit: Cryptography foundations and the solving workflow.
Worked-case data: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

answer: Otherwise one ciphertext symbol can refer to multiple plaintext letters, so a unique inverse does not exist. (4 points).

12. (4 points) Why state the base?

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

answer: Place value, allowable digits, and the carry multiplier depend on the base. (4 points).

13. (2 points) Define “Caesar cipher” in the context of Codebusters.

answer: A fixed alphabet shift. (2 points).

14. (4 points) What can an unused Baconian group indicate?

Unit: Baconian and fractionated Morse systems.
Worked-case data: Decode AAAAA AAAAB AABAA under the modern 26-letter Baconian convention.

answer: Wrong class assignment, incorrect grouping, transcription error, or a different alphabet convention. (4 points).

15. (2 points) Define “Transposition” in the context of Codebusters.

answer: Rearrangement of letters without changing their identities. (2 points).

16. (4 points) What should a solver write for K1/K2/K3?

answer: Explicit plaintext and ciphertext alphabets, keyword deduplication, remaining-letter order, and any relative shift stated in the problem. (4 points).

17. (4 points) List all checks on a cryptarithm candidate.

Unit: Xenocrypts and cryptarithms.
Worked-case data: Verify the candidate SEND = 9567, MORE = 1085, MONEY = 10652.

answer: Full arithmetic, consistent repeated letters, distinct digits when required, correct base, and no forbidden leading zeros. (4 points).

18. (6 points) Solve this case and explain your reasoning: A = 0 and a Caesar key of +3 are used to encode CAT. Decode FDW and verify.

reasoning: CAT corresponds to 2, 0, 19. Add 3 modulo 26 to obtain 5, 3, 22: FDW.
Decode FDW by subtracting 3 modulo 26: 2, 0, 19.
Re-encrypt CAT and compare all three symbols with FDW. (3 points).

conclusion: The recovered plaintext is CAT, and independent re-encryption confirms the key direction. (3 points).